

Lab Session 03

Carry out user view and structural view analysis for the suggested system: Use Case and Class Diagrams

Use case diagram

Use case diagrams are used to describe a set of actions (**use cases**) that some system or systems (**subject**) should or can perform in collaboration with one or more **external users** of the system (**actors**). Each use case should provide some observable and valuable result to the actors or other stakeholders of the system.

(System) Use case diagrams are used to specify:

- (external) requirements, required usages of a system under design or analysis (subject) - to capture what the system is

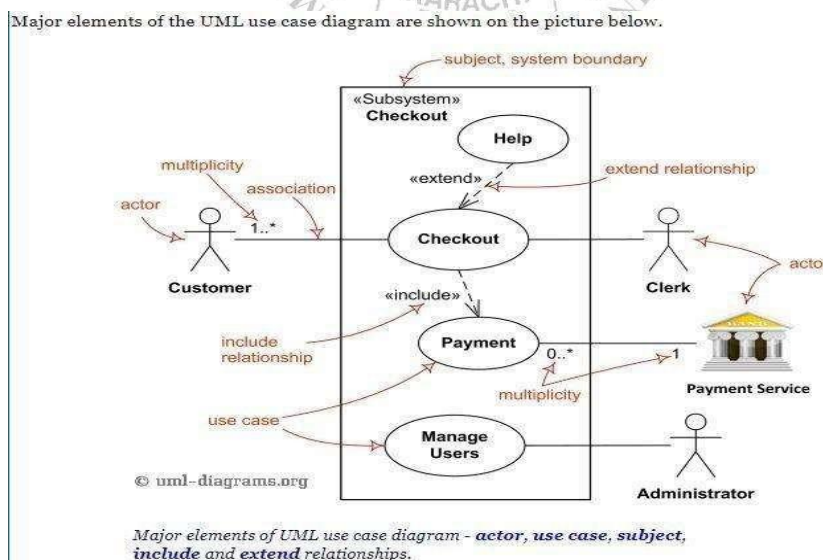
Create Use Case Diagram

To create a Use Case Diagram in StarUML:

Select first an element where a new Use Case Diagram to be contained as a child.

Select **Model | Add Diagram | Use Case Diagram** in Menu Bar or select **Add Diagram | Use Case Diagram** in Context Menu.

To create use case subjects, actors, use cases, include and extend relationship etc., select the desired option from toolbox and drag on the main diagram. Upon double clicking, all the respective options will be made available.



the functionality offered by a subject – what the system can do;

Requirements the specified subject poses on its environment - by defining how environment should interact with the subject so that it will be able to perform its services.

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Class diagram

Class diagram is **UML structure diagram** which shows structure of the designed system at the level of **classes** and **interfaces**, shows their **features**, **constraints** and relationships **associations**, **generalizations**, **dependencies**, etc.

Create Class Diagram

To create a Class Diagram in StarUML:

To create classes, enumerations, association, composition, generalization and composition relationships, select the respective option from the toolbox.

First select an element where a new Class Diagram to be contained as a child.

Select **Model | Add Diagram | Class Diagram** in the Menu Bar or select **Add Diagram** Context Menu.

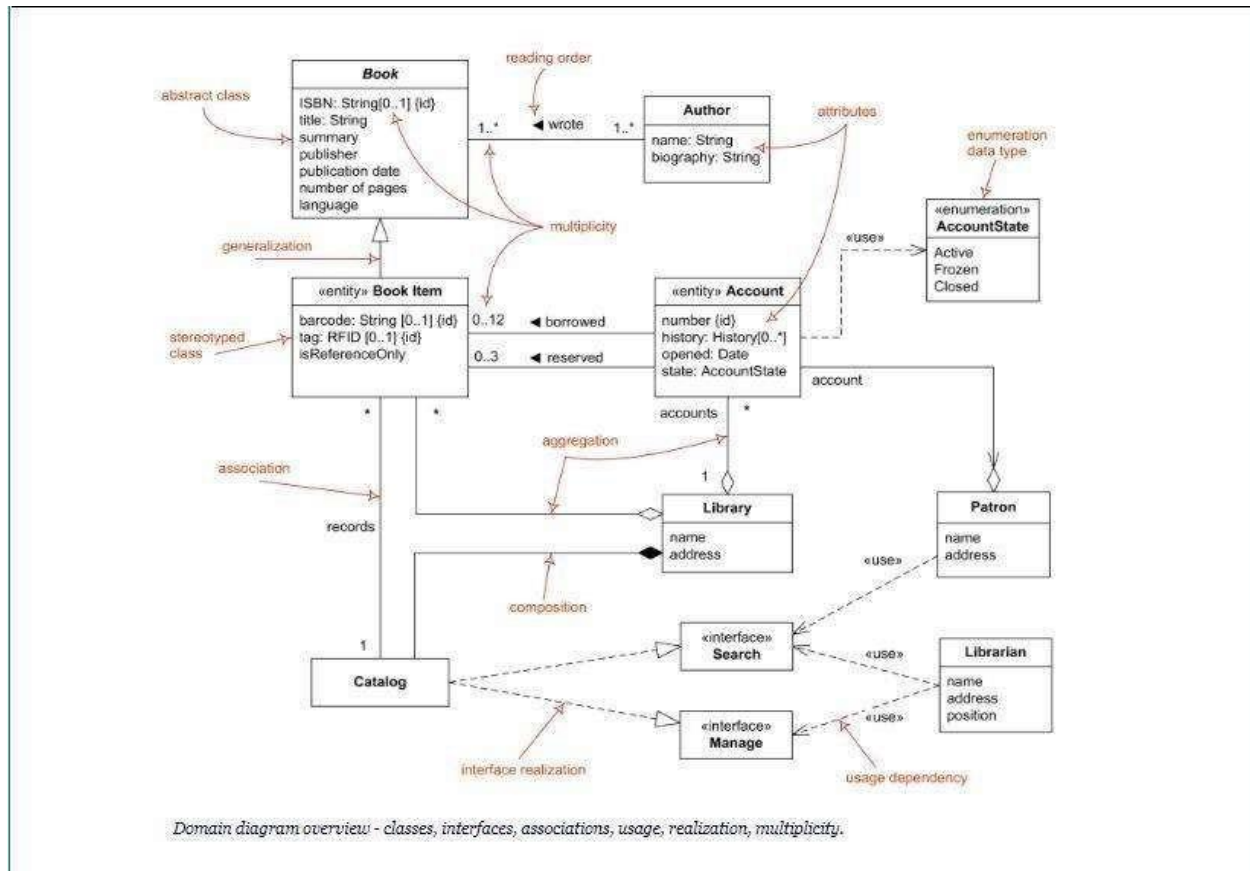


Fig 3.2 Major Elements of UML Class Diagram

Online Shopping System

Web Customer actor uses some web site to make purchases online. Top level **use cases** are **View Items**, **Make Purchase** and **Client Register**. View Items use case could be used by customer as top level use case if customer only wants to find and see some products. This use case could also be used as a part of Make Purchase use case. Client Register use case allows customer to register on the web site, for example to get some coupons or be invited to private sales. Note, that **Checkout** use case is **included use case** not available by itself - checkout is part of making purchase.

View Items use case is **extended** by several optional use cases - customer may search for items, browse catalog, view items recommended for him/her, add items to shopping cart or wish list. All these use cases are extending use cases because they provide some optional functions allowing customer to find item.

Customer Authentication use case is **included** in **View Recommended Items** and **Add to Wish List** because both require the customer to be authenticated. At the same time, item could be added to the shopping cart without user authentication.

UML Use Case Diagram Example

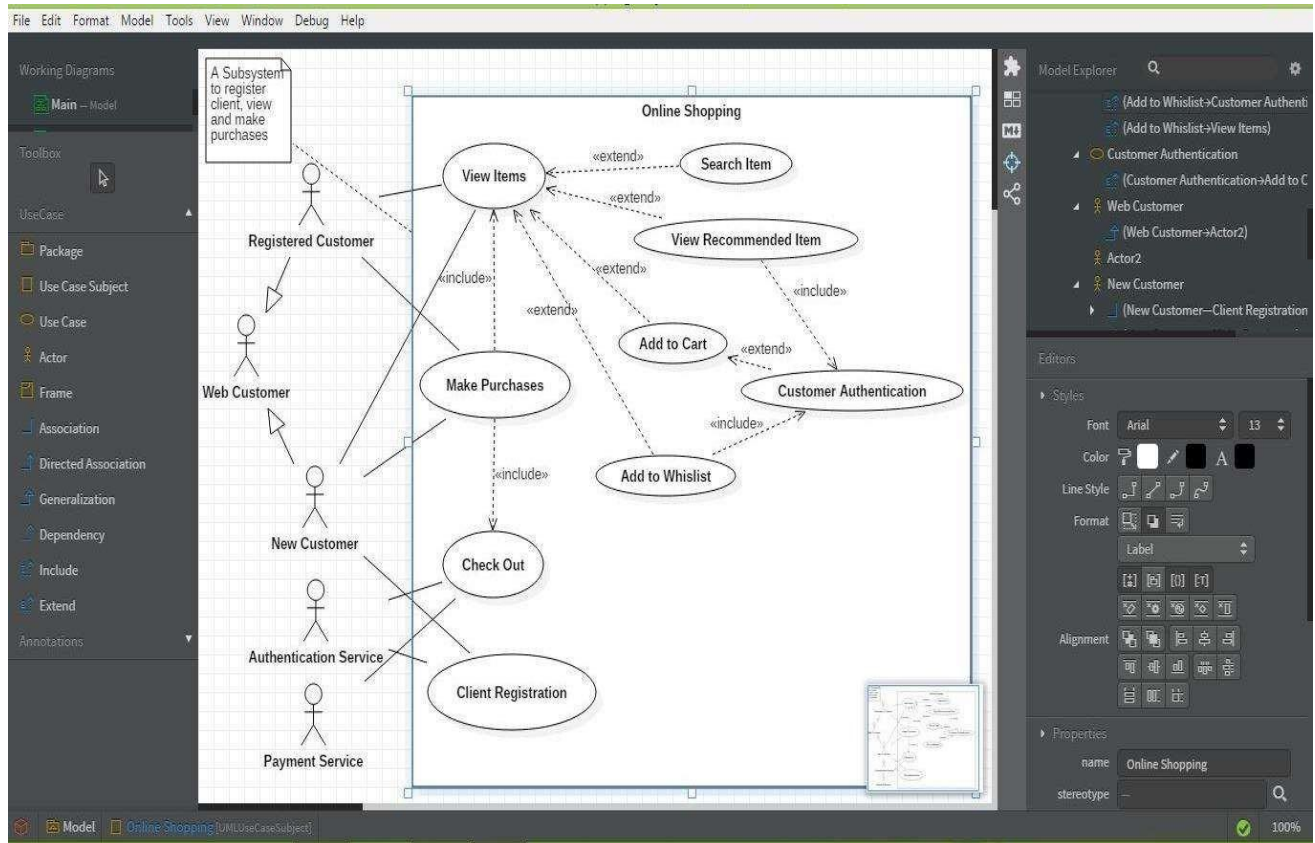
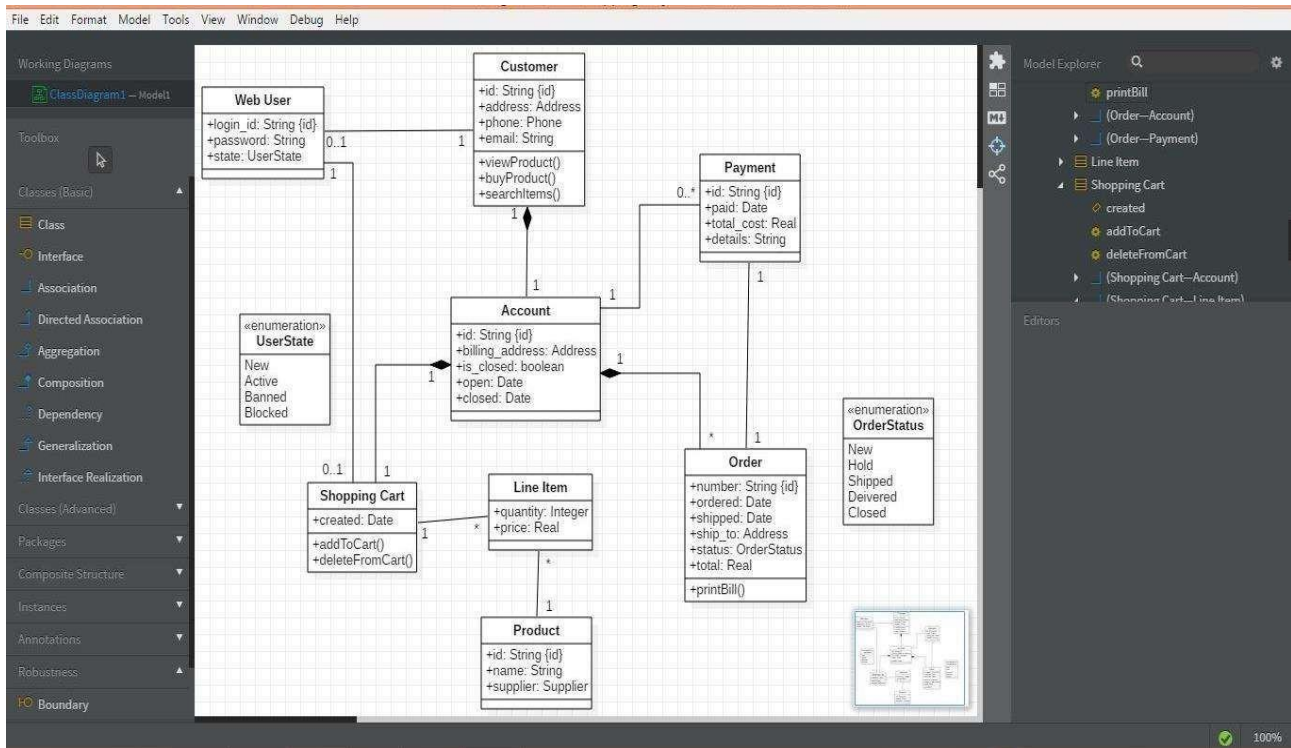


Fig 3.3 UML Use Case Diagram for Online Shopping System

UML Class Diagram Example



EXERCISES:

1. Construct use case and class diagram for an automated teller machine (ATM) or the automatic banking machine (ABM). Customer uses a bank ATM to check balances of his/her bank accounts, deposit funds, withdraw cash and/or transfer funds (use cases). ATM Technician provides maintenance and repairs to the ATM. Also attach printout.
2. For a hypothetical “Food Ordering App”, identify possible classes, attributes and their operations. Also show the relationships among various classes using StarUML. Attach its printout.
3. In “Online Shopping System”, Check Out use case includes several required uses cases. Web customer should be authenticated. It could be done through user login page, user authentication cookie ("Remember me"). Web site authentication service is used in all these use cases. Checkout use case also includes Payment use case which could be done by using external credit payment service. Extend the given use case diagram incorporating the given information. Also attach the

1. ATM / ABM System

Use Case: Customer performs transactions; Technician does maintenance; Bank Server verifies all financial operations.

Class Diagram: ATM interfaces with Customer and Transaction; Account handles credit/debit; CardReader manages cards; Bank verifies transactions; Customer owns accounts; Transaction links to Account.

2. Food Ordering App

Classes & Attributes: User manages profile and authentication; Order calculates total and tracks status; Restaurant maintains menu; FoodItem represents menu items; Payment processes transactions.

Relationships: Restaurant aggregates FoodItems; User has multiple Orders; Order contains multiple FoodItems; Order associates with Payment.

3. Online Shopping System

Use Case: Check Out includes Authenticate User and Payment; Authenticate User generalizes into User Login Page or Authentication Cookie; Web Customer interacts with Check Out; external services act as actors.

